

LOLLAPALOOZA CHICAGO

*A Data-Driven Analysis of Artist Popularity,
Genre Evolution & Chicago's Events Landscape*

Research Overview

CONTEXT

- Lollapalooza has run annually in Grant Park since 2005
- Estimated \$500M annual economic impact for Chicago
- 20 festivals · 150+ performers per year · 8+ stages
- Streaming era (Spotify ~2010s) changed how music is consumed globally

HYPOTHESIS: Lollapalooza has become more mainstream over time.

- **Popularity** ↑
Artists have higher Spotify popularity over time
- **Genres diversify**
More rap + electronic alongside rock/indie
- **Economic impact**
Influences Chicago economy (varies by sector)

DATASETS: Lollapalooza Performers 2005–2025 (Kaggle / Wikipedia) · City of Chicago Special Events (Data Portal) · Chicago Building Permits (833K rows)

The Data at a Glance

2,841

Performers

20 festivals · 2005–2025

~2,253

Unique Artists

across all editions

53 / 100

Avg Popularity

std dev: 20.5

593

Special Events

21 event types in Chicago

49.7%

Genre Missing

1,414 of 2,841 rows

196

Headliners

93.1% flag missing

$R^2 =$

0.21

Regression Score

MAE = 13.56 pts

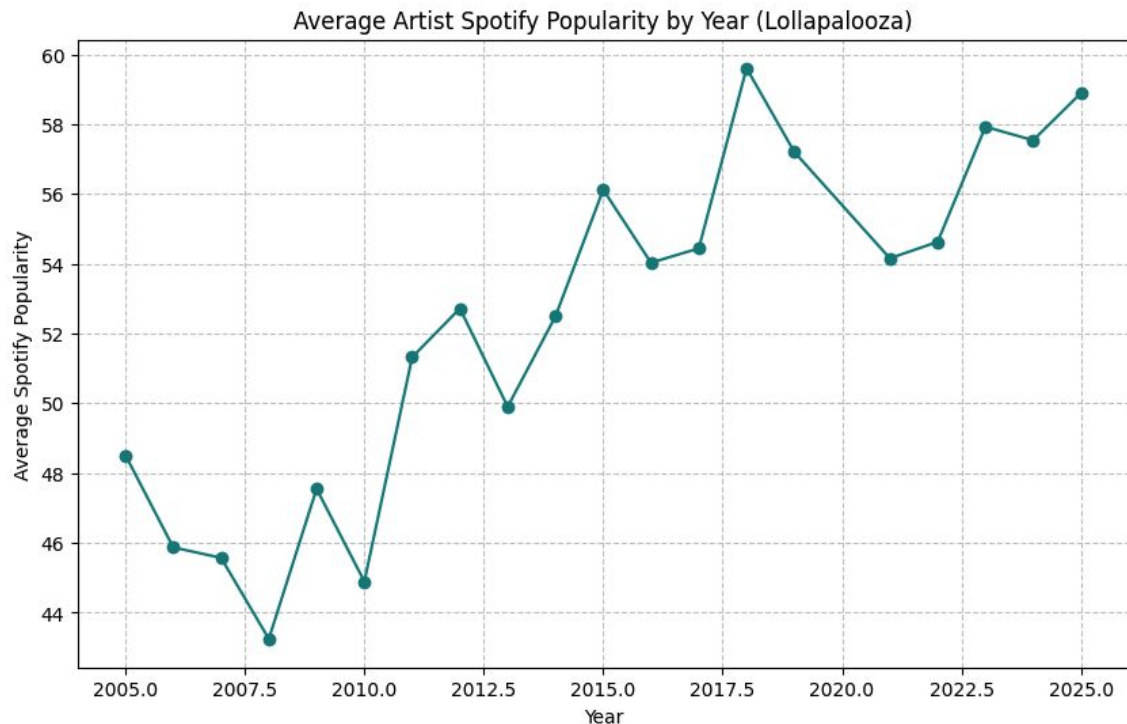
38%

Block Parties

225 of 593 Chicago events

Key missingness: headliner flag dropped from regression (93.1% null); genre imputed with 'Unknown' for modeling.

Popularity Has Risen Over 20 Years



KEY FINDINGS

~13–15 pt

increase over 20 years

44–50

avg in 2005–10

57–63

avg in 2018–25

Acceleration

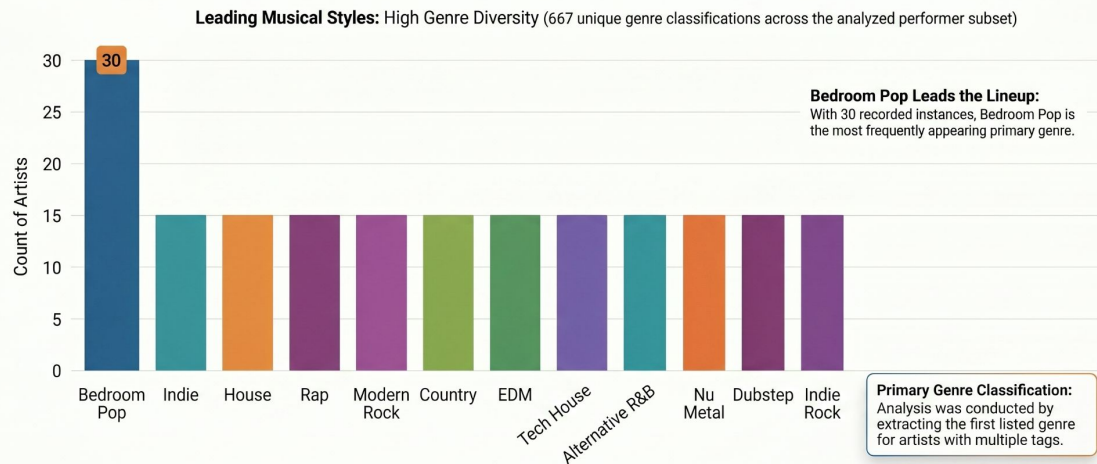
after 2017 — Spotify hits 400M users

Note: popularity scores reflect April 2025 values — earlier years may be slightly overstated due to measurement timing.

Genre Diversification

Figure 2. Genre Frequency Among Lollapalooza Performers (n=1,427)

This analysis utilizes a dataset of 2,841 Lollapalooza performers. To ensure accuracy in genre representation, the data was filtered to include only the 1,427 artists with recorded genre information, removing the 1,414 entries where genre data was missing.



NotebookLM

INSIGHT

Bedroom pop

— a 2010s streaming-native genre — ranks #1

Rap & EDM

now present; atypical in 2005–08 lineups

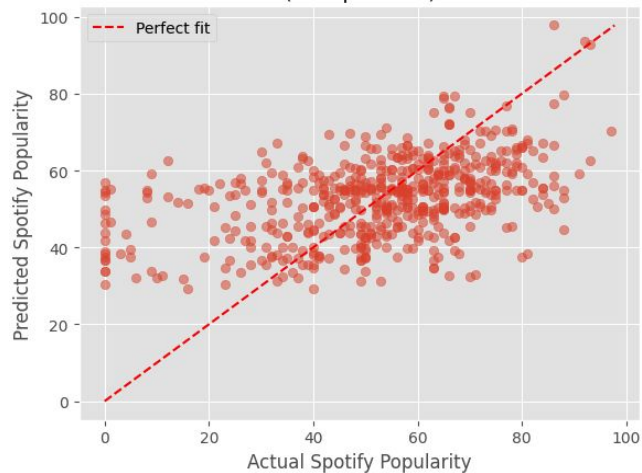
Rock & indie persist

— not displaced, but sharing the lineup with streaming-dominant genres

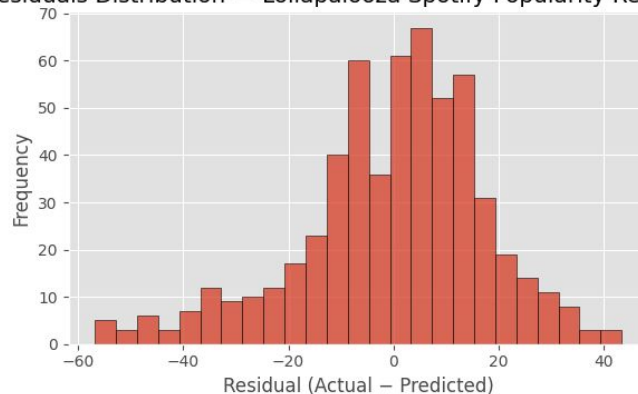
49.7% of rows missing genre — analysis based on 1,427 non-null rows.

What Predicts Artist Popularity?

Linear Regression: Actual vs Predicted Spotify Popularity (Lollapalooza)



Residuals Distribution — Lollapalooza Spotify Popularity Regression



Test $R^2 = 0.2068$ · Train $R^2 = 0.2442$

MAE = 13.56 pts · RMSE = 17.65

70 features · 80/20 train-test split

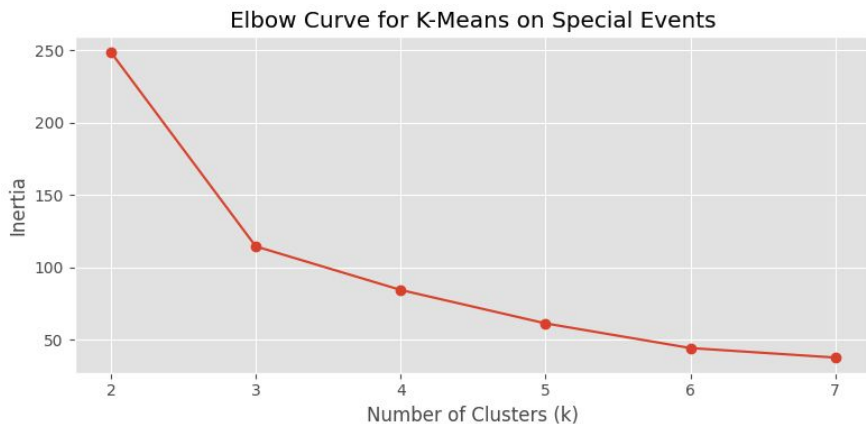
SBC West stage: +32.4 · Rap genre: +28.4

SBC East: +27.0 · Red Bull Soundstage: +25.2

Garage rock: -10.4 · Alt R&B: -5.8

$R^2 \sim 0.21$ is typical for music popularity data

Chicago's Event Landscape



K-MEANS CLUSTERING (k=4)

Cluster 0

69 events · 31.8%

Core summer window (May–Aug)

Cluster 1

52 events · 24.0%

Spring / early summer

Cluster 2

59 events · 27.2%

Winter / early-year events

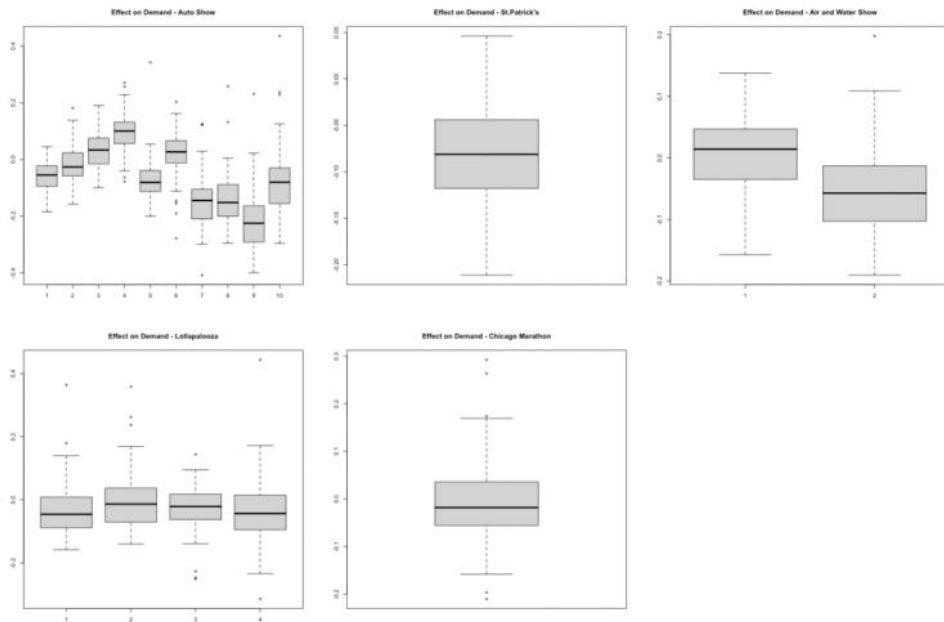
Cluster 3

37 events · 17.1%

Late-fall / holiday (e.g. Christkindlmarket)

Block Parties = 38% of all 593 permitted events. Lollapalooza is the only event combining multi-day + 150+ performers + multi-stage Grant Park footprint.

Economic Impact: Taxi Demand



FINDING

Lollapalooza effect on taxi demand is near zero

Large events don't always increase all economic activity

Likely explanation: attendees use walking, transit, or rideshare — not taxis

Broader takeaway: vendors, hotels & restaurants benefit more directly than taxi services

Based on Bayesian Structural Time Series (BSTS) model of Chicago taxi trips 2013–2023. Analysis by Lea Wu (2024), Towards Data Science.

Interactive Chatbot

What it uses

- NotebookLM with uploaded datasets + analysis
- Lollapalooza data (2,841 performers)
- Chicago events data (593 events)
- My report + findings

How it works

- Uses **natural language queries** (no fixed keywords)
- Retrieves answers directly from uploaded content
- Generates responses based on my analysis

What it can answer

- Popularity trends over time
- Genre changes
- Headliners vs non-headliners
- Chicago events + real-world impact

Why this meets requirements

- Uses project data as input
- Interactive chat-based design
- Supports multiple question types
- More flexible than rule-based chatbot

Demo

The screenshot displays the Google NotebookLM interface. At the top, the notebook title is "Lollapalooza Performance History and Artist Popularity Data". The interface is divided into three main sections: Sources, Chat, and Studio.

Sources: This section on the left lists the data sources used in the notebook. It includes a search bar and a list of sources with checkboxes for selection:

- Special_Events.csv
- data project - milestone 1-1.pdf
- data project - milestone 2-1.pdf
- lollapalooza.csv

Chat: The central section shows the notebook's content. It features a guitar icon and the title "Lollapalooza Performance History and Artist Popularity Data". Below the title, it states "4 sources · Apr 28, 2026". The main text describes a data science workflow using Python to analyze and predict trends in Chicago city records, specifically focusing on building permit data. It mentions the use of visualizations like histograms and time series plots, and the Prophet library for forecasting. The text concludes by stating that the source serves as a technical guide for performing regressional and time series analysis on public datasets.

Studio: The right section contains various interactive tools and visualizations. It includes a grid of cards for "Audio Overview", "Video Overview", "Reports", "Quiz", "Data Table", "Slide Deck", "Mind Map", "Flashcards", and "Infographic". Below this grid, there are two featured visualizations:

- Music Festival Genre Frequency Chart (4 sources · 1h ago)
- Music Festival Artist Popularity Trends (4 sources · 2h ago)

At the bottom of the Chat section, there are three prompts for further exploration:

- How are missing values and data outliers handled in this project?
- Describe the seasonal trends found in Chicago's building permit applications.
- How does the project model relationships between building and cafe permits?

<https://notebooklm.google.com/notebook/8b35cceb-a3ad-44d5-8d07-dcbac0c56a11>

Conclusions

Primary hypothesis **CONFIRMED**

Avg Spotify popularity increased ~13–15 pts from 2005–2025; strongest growth post-2017

Genre hypothesis **PARTIALLY CONFIRMED**

Bedroom pop, rap, electronic have grown; rock/indie persist and were not displaced

Regression insight

Stage (+32.4 SBC West) and genre (+28.4 rap) are strongest predictors; $R^2=0.21$ typical for music data

Taxi demand finding

Lollapalooza has near-zero effect on taxi trips — economic impact is sector-specific (hotels, vendors, restaurants)

Chicago event landscape

Block Parties dominate (38%); Lollapalooza is categorically unique — multi-day, 150+ performers, Grant Park